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Pharmacology of the Cardiovascular System

Hyperlipidaemia

Ken Wojcikowski

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Pharmacology of the Cardiovascular System: Hyperlipidaemia

COMMONWEALTH OF AUSTRALIA

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Medications of the Cardiovascular System: Medications for Altering Lipid Levels

Lesson 1:

Understanding Lipids

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Objectives for the first three lessons

- Define the terms hyperlipidaemia, dyslipidaemia, HDL and LDL
- Describe atherosclerosis, including the risk factors, and potential consequences of plaques on the arteries

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Lipids

Tri(acyl)glycerides
Used in energy metabolism
ONE glycerol and **three fatty acids**

Phospholipids
Important structural constituents of lipoproteins, blood- clotting components, the myelin sheath, and cell membranes

Cholesterol
Chemical activity similar to other lipid substances

Hyperlipidaemia
Elevated levels of one or all of the above

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Types of Lipoproteins; Classified by Density

- **Chylomicrons** (transfer lipids from GIT to liver)
- **Very–low-density lipoprotein** (VLDL)
  - Carries large amounts of triglycerides
- **Intermediate-density lipoprotein** (IDL)
- **Low-density lipoprotein** (LDL)
  - Main carrier of cholesterol from the liver to the cells
- **High-density lipoprotein** (HDL)
  - The good protein

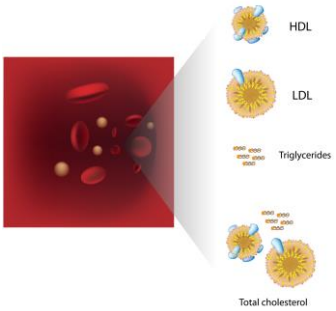
Schematic drawing of HDL, a protein



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The common lipid profile screening includes:

- **HDL: a protein**
- **LDL: a protein**
- **Triglycerides: a type of fat**
- **Total Cholesterol: both HDL and LDL**



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Dyslipidaemia/hyperlipidaemia

Virtually synonymous

- **High levels of lipids in the blood (but dyslipidaemia may also describe low amounts of HDL)**
  - **Secondary hyperlipidaemia (caused by multiple risk factors)**
  - **Primary/familial hypercholesterolemia**
  - **Primary/familial hypertriglyceridemia**

Lipid	NORMAL Blood level
LDL (Bad)	<2.0 mmol/L
HDL (Good)	>1.0 mmol/L
Cholesterol	<4.0 mmol/L
Triglycerides	<1.5 mmol/L

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Medications of the Cardiovascular System:
Medications for Altering Lipid Levels
Lesson 2:
Understanding Atherosclerosis

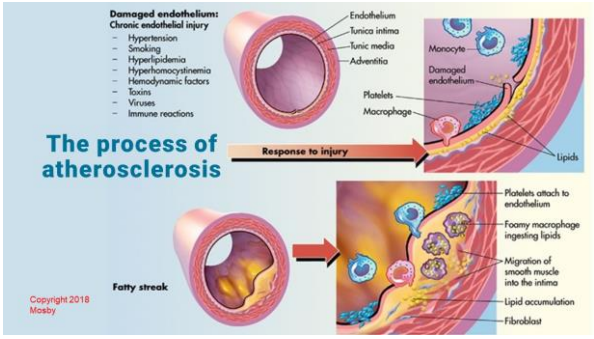
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Xanthelasma



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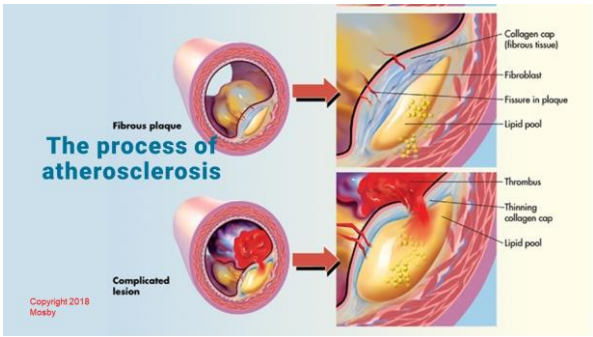
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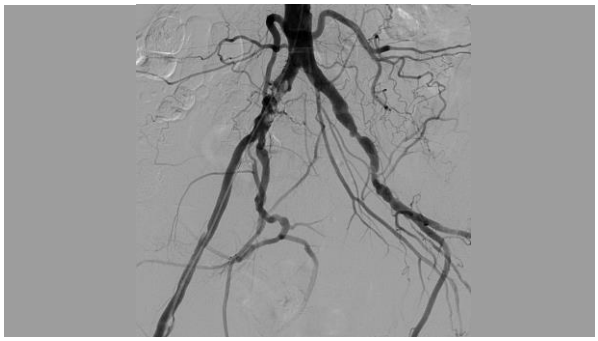
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Medications of the Cardiovascular System:
Medications for Altering Lipid Levels
Lesson 3:
Understanding Atherosclerosis (cont.)

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Atherosclerosis of the Coronary Artery



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Hyperlipidaemia Medications

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Objective the remaining lessons

- Identify and describe the therapeutic actions and major side effects of common pharmacological agents used in the treatment of dyslipidaemia or hyperlipidaemia.

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Medications of the Cardiovascular System:

Medications for Altering Lipid Levels

Lesson 4:

HMG-CoA reductase inhibitors (the statins)

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HMG-CoA reductase inhibitors (the statins)

- All end in the suffix …statin
  - Atorvastatin (Lipitor),
  - Simvastatin (Zocor, Simvar)
  - Lovastatin (Mevacor)

HMG-CoA reductase is the enzyme that the liver uses to make cholesterol

- BONUS!: They also have anti-inflammatory and antithrombotic effects, stabilising plaques and preventing their rupture and increase longevity

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HMG-CoA reductase inhibitors (the statins)

Adverse effects:
- GI disturbances,
- Possible rhabdomyolysis (rare, but may cause acute renal failure, so be aware of this!)
  - Tea coloured urine and muscle soreness are the first signs

Contraindications:
- Pregnancy Category D
- Liver disease



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HMG-CoA reductase inhibitors (the statins)

Check recent evidence before prescribing one statin over the other

These do vary greatly in efficacy and the one that is associated with the greatest increase in longevity in recent studies should be chosen (unless the safety profile makes it inappropriate)!

Not to be used by women of reproductive years
Do not use in liver damage
Monitor Liver Function Test (LFT) in all patients
Raised AST and ALT means liver damage

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Medications of the Cardiovascular System:
Medications for Altering Lipid Levels
Lesson 5:
Medications that affect the absorption and excretion of cholesterol

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Cholesterol absorption inhibitors

- Cholesterol absorption inhibitors (relatively new)
  - Ezetimibe (Ezetrol) eh zet im ib
    - Limit absorption of cholesterol in the small intestines
    - Liver increases uptake of systemic cholesterol
    - Adverse effects: GI disturbances
- Unlike most other combinations of lipid normalising medications, Ezetimibe can be combined with statins
  - Most other combinations of lipid lowering drugs are contraindicated



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Bile acid sequestrants

Cholestyramine (Questran) chole steera meen

- Like Styrofoam – sucks up the bile salts in the small intestines
  - Bile salts excreted instead of reused

☹ The initial decreases in LDL may be negated by the liver
☹ The medication tastes terrible
☹ The medication binds fat soluble vitamins, D, K, E and A
☹ The medication causes constipation

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Bile acid sequestrants

Cholestyramine (Questran)

Contraindications:
- **pregnancy (decrease the absorption of fats and fat soluble vitamins; babies need fats and fat soluble vitamins);**
- **do not consume with any drugs as it may decrease absorption**

Good news – it is useful in decreasing the diarrhea associated with Crohn's disease and along with antibiotic therapy in treatment of GIT Clostridium infections (binds the toxins and decreases the diarrhea)

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Medications of the Cardiovascular System:
Medications for Altering Lipid Levels
Lesson 6:
Nicotinic Acid

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Nicotinic acid



Vitamin B3; (Niaspan)
- Huge dose of nicotinic acid (1-3 grams daily)
- Decreases formation of free fatty acids in adipose tissue
- Not suitable in all types of hyperlipidaemia, but can decrease triglycerides by 25%
- Decreases LDL and increases HDL by 30%

Not to be used in
- Liver disease, Diabetes mellitus,
- Gout or Stomach ulcers

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Nicotinic acid

Adverse effects include flushing of the ears and face, pruritus, GIT ulcers, nausea, diarrhoea, heartburn, and possible hypotension

- May need to start off with small dose to reduce flushing
- To reduce flushing: Titrate dose, take just after meals, or suggest an aspirin 30 minutes prior

Nicotinic acid not to be taken with

- Statins
- Fibrates



Flushing/ (burning feeling) of the ears is a common side effect

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Medications of the Cardiovascular System:
Medications for Altering Lipid Levels
Lesson 7:
Fibrates

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Fibrates

Largely superseded by statins and other newer medicines

Increase the breakdown of lipids in tissues

- Fenofibrate (Lipidil; the example to know)

LDL sometimes rises but is partially offset by increases in HDL, an inhibition of platelet aggregation and an increase in fibrinolysis



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Fibrates

Adverse effects:

- Abdominal pain, nausea, diarrhoea
- Myalgia (muscle pain) may be the first symptom of rhabdomyolysis
- Gall stones

Individuals should be advised to speak to their doctor if they experience muscle pain, weakness and tenderness as these signs may be indicative of myalgia rhabdomyolysis, not myalgia)

Contraindicated in those taking:

- Warfarin
- Statins
- Nicotinic Acid

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Medications of the Cardiovascular System:
Medications for Altering Lipid Levels
Lesson 8:
Special notes about lipid lowering drugs

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Special notes about lipid lowering drugs

- Always 'monitor the liver' when giving hypercholesterolaemic drugs
  - How? The Liver Function Test (LFT, specific enzymes AST, ALT)
- Most combinations of lipid lowering drugs are contraindicated
  - Exception to the rule: Ezetimibe can be combined with statins
- None of the lipid normalizing drugs are ideal during pregnancy.
  - At one time statins were thought to be teratogenic but are now listed as pregnancy category D
  - In addition, babies need cholesterol !!!!!!!!!!
  - Dietary approaches are more suitable than drugs during pregnancy if that is possible (those with familial hypercholesterolemia do not respond to lifestyle changes very well).

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